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Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

AI-Based Compliance & Audit Automation Platform

AN INTERNSHIP REPORT

Submitted by

GIRISHA S B – 20221CCS0135

Under the guidance of,

Dr. DEEPTHI S

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (Cyber Security)

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APRIL 2026

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BONAFIDE CERTIFICATE

Certified that this report "AI-Based Compliance and Audit Automation Platform " is a bonafide work of GIRISHA S B (20221CCS0135), who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING (SPECIALIZATION DIVISION), CYBER SECURITY during 2025-2026.

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DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING (Cyber Security), at Presidency University, Bengaluru, named Girisha S B, hereby declare that the internship work titled " AI-Based Compliance and Audit Automation Platform " has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING (Spl. Division) during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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INTERNSHIP OFFER LETTER

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ISO 9001:2015 Certified

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Internship Offer Letter

HackerHub8 LLP |
CIN – ACR-1466

Date: 15 Dec 2025

Dear **GIRISHA S B**,

We are pleased to offer you an internship position at **HackerHub8 LLP**. This letter confirms your selection for the role of **Internship Role – Jr. Cyber Security Intern** for the duration **from Start** 04 Jan 2026 **to** 01 May 2026.

Internship Details

Role: Jr. Cyber Security Intern
Department: Cybersecurity
Mode: Online
Working Hours: Flexible (Updated on Mail)

Your Tasks & Responsibilities Include:

- Participating in daily/weekly live sessions, labs, and training modules
- Working on assigned projects, tasks, or assessments
- Reporting progress to the allocated mentor
- Following organizational guidelines and maintaining confidentiality at all times

Terms & Conditions

You are expected to conduct yourself professionally throughout the internship period and uphold the values and policies of HackerHub8 LLP . Any form of misconduct or policy violation may lead to immediate termination of the internship.

Certification

Upon successful completion of your internship and assigned tasks, you will receive an **Internship Completion Certificate** from HackerHub8 LLP.
We are excited to welcome you as part of our team and look forward to your valuable contributions.
Kindly confirm your acceptance of this offer to proceed with the onboarding process.

Welcome to HackerHub8 LLP Team
Warm Regards,
HR Manager | Ravi G

HACKERHUB8 LLP SINCE 2019 KARNATAKA

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ACKNOWLEDGEMENTS

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved **Chancellor**, **Pro-Vice Chancellor**, and **Registrar** for their support and encouragement in completion of the internship.

I would like to sincerely thank my internal guide **Dr. Deepthi S**, Assistant Professor, Presidency School of Computer Science and Engineering (Spl. Division), Presidency University, for his moral support, motivation, timely guidance, and encouragement provided to me during the period of internship work.

I am also thankful to **Dr. Anandaraj S.P**, Professor, Head of the Department, Presidency School of Computer Science and Engineering (Spl. Division), Presidency University, for his mentorship and encouragement.

I express my cordial thanks to **Dr. Shakkeera L**, Associate Dean, Presidency School of Computer Science and Engineering (Spl. Division), Presidency University, for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

I am grateful to **Dr Geetha A,** School Internship Coordinator and Program Internship Coordinator, Presidency School of Information Science and Engineering, for facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance

I am also grateful to the Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

I extend my sincere gratitude to **HackerHub8 LLP**, Bengaluru, for providing me with the opportunity to work as a Jr. Cyber Security Intern and for their continuous guidance and support throughout the internship period.

Girisha S B

ABSTRACT

The increasing complexity of cybersecurity compliance frameworks such as ISO 27001, SOC 2, HIPAA, and DPDP has made manual compliance management a challenging and time-consuming task for organizations. Traditional approaches rely heavily on spreadsheets and manual audits, leading to inefficiencies, delayed remediation, and lack of real-time visibility into compliance posture. This creates a critical need for an intelligent, automated system that can streamline compliance tracking, identify gaps, and improve audit readiness.

This internship project focuses on the design and development of an AI-Based Compliance and Audit Automation Platform at HackerHub8 LLP. The system automates the entire compliance lifecycle, including control tracking, compliance scoring, gap detection, and audit report generation. A weighted scoring engine is implemented to calculate the overall compliance score based on control implementation status, where implemented controls contribute full weight, partially implemented controls contribute partial weight, and non-implemented controls contribute zero weight.

The platform incorporates a rule-based AI gap detection engine that analyzes control statuses across multiple frameworks and automatically identifies compliance gaps. Each gap is classified into risk levels such as High, Medium, and Low, and appropriate remediation strategies are generated dynamically. The system supports four major compliance frameworks, consisting of 124 controls, providing a realistic and scalable compliance environment.

The system is built using a modern full-stack architecture with Python FastAPI as the backend, React.js as the frontend, and SQLite/PostgreSQL as the database. The platform features a dynamic dashboard displaying key performance indicators such as compliance percentage, control status distribution, and framework-wise analysis. Additionally, it provides one-click automated audit report generation in PDF format, enabling organizations to prepare audit-ready documentation efficiently.

The system was evaluated using a simulated dataset containing hundreds of control status records across multiple frameworks. The results demonstrate that the platform significantly reduces manual effort, improves accuracy in compliance assessment, and provides actionable insights for faster remediation. Overall, the proposed system offers an efficient, scalable, and intelligent solution for modern compliance management and audit automation

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ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
JWT	JSON Web Token
KPI	Key Performance Indicator
ORM	Object Relational Mapper
REST	Representational State Transfer
SQL	Structured Query Language
DB	Database
UI	User Interface
HTTP	Hypertext Transfer Protocol
CPU	Central Processing Unit
RAM	Random Access Memory
ISO	International Organization for Standardization
SOC	System and Organization Controls
HIPAA	Health Insurance Portability and Accountability Act
DPDP	Digital Personal Data Protection
GRC	Governance, Risk, and Compliance
PDF	Portable Document Format

Chapter – 1

INTRODUCTION

1.1 Background

The rapid digital transformation of organizations across industries has led to a significant increase in the adoption of information systems and cloud-based infrastructures. With this growth, ensuring compliance with cybersecurity standards such as ISO 27001, SOC 2, HIPAA, and DPDP has become a critical requirement. These frameworks define structured guidelines to protect sensitive data, ensure privacy, and maintain system integrity.

However, managing compliance across multiple frameworks is a complex and resource-intensive task. Organizations are required to track hundreds of controls, monitor their implementation status, maintain evidence, and prepare for periodic audits. Traditionally, this process is performed manually using spreadsheets, documents, and checklists, which often results in inefficiencies, inconsistencies, and lack of real-time visibility.

Manual compliance management also introduces challenges such as delayed identification of gaps, inaccurate reporting, and increased audit preparation time. As organizations scale, these issues become more pronounced, leading to higher operational costs and increased risk of non-compliance.

Furthermore, the integration of intelligent technologies such as artificial intelligence and data analytics has enhanced the effectiveness of compliance management systems. These technologies enable automated analysis of control data, early detection of compliance gaps, and generation of actionable insights. As a result, organizations can address issues during audits.

Therefore, there is a strong need for a unified platform that can manage multiple compliance frameworks, provide real-time visibility, and automate critical processes such as scoring, gap detection, and reporting. The proposed system aims to address these challenges by offering a scalable and efficient solution for modern compliance management. The growing number of regulations and frequent updates to standards make it difficult to maintain compliance using traditional methods.

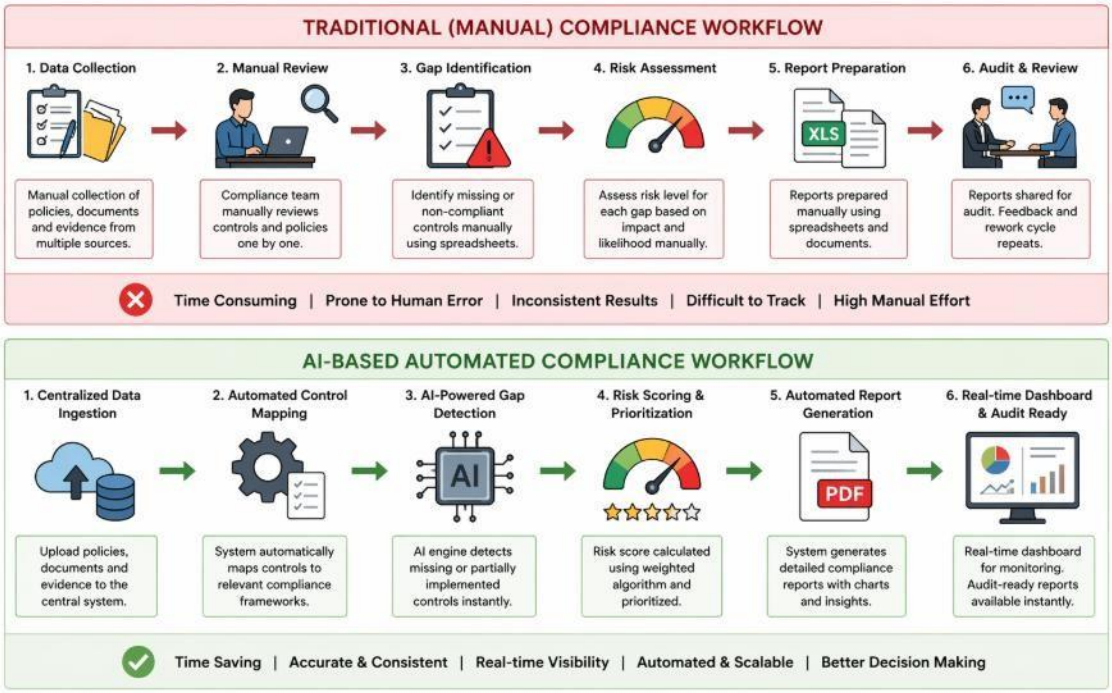

Figure 1.1: Traditional vs Automated Compliance Workflow

1.2 Statistics of Internship

The internship was carried out at HackerHub8 LLP, a cybersecurity-focused organization, where the primary objective was to design and develop an automated compliance management platform. The internship provided hands-on experience in full-stack development, cybersecurity compliance, and AI-based system design.

Table 1.1: Statistics of the Internship

Parameter	Details
Internship Title	AI-Based Compliance & Audit Automation Platform
Organization	HackerHub8 LLP, Bengaluru, Karnataka
Department	Cybersecurity
Mode	Online

Duration	04 January 2026 to 01 May 2026
Domain	Cybersecurity, Compliance Automation
Primary Tech Stack	Python, FastAPI, React, SQLite/PostgreSQL
Main Output	Full-stack compliance automation system

1.3 Prior Existing Technologies

Several tools and platforms exist for compliance management and audit preparation. These include manual tracking systems, enterprise governance tools, and security compliance platforms. However, most of these systems either lack automation or are complex and expensive to implement.

Table 1.2: Existing Compliance Systems

Category	Tools/Systems	Limitation
Manual Systems	Excel, Google Sheets	Time-consuming, error-prone
GRC Platforms	ServiceNow, Archer	Expensive, complex
Compliance Tools	Drata, Vanta	Limited customization
Audit Tools	AuditBoard	Focused only on reporting
Category	Systems/Tools	Strength

1.4 Proposed Approach

To overcome the limitations of traditional systems, this project proposes an AI-Based Compliance and Audit Automation Platform that integrates automation, scoring, and intelligent analysis into a single system.The system introduces a weighted compliance scoring engine, which calculates the overall compliance score based on control implementation status.

- Implemented → 100% weight

- Partially Implemented → 50% weight
- Not Implemented → 0% weight

In addition, a rule-based AI gap detection engine analyzes control data and identifies missing or weak controls. Each gap is categorized into:

High Risk

Medium Risk

Low Risk

The system also generates automated remediation suggestions and provides a one-click audit report generation feature, making the entire compliance lifecycle efficient and streamlined.

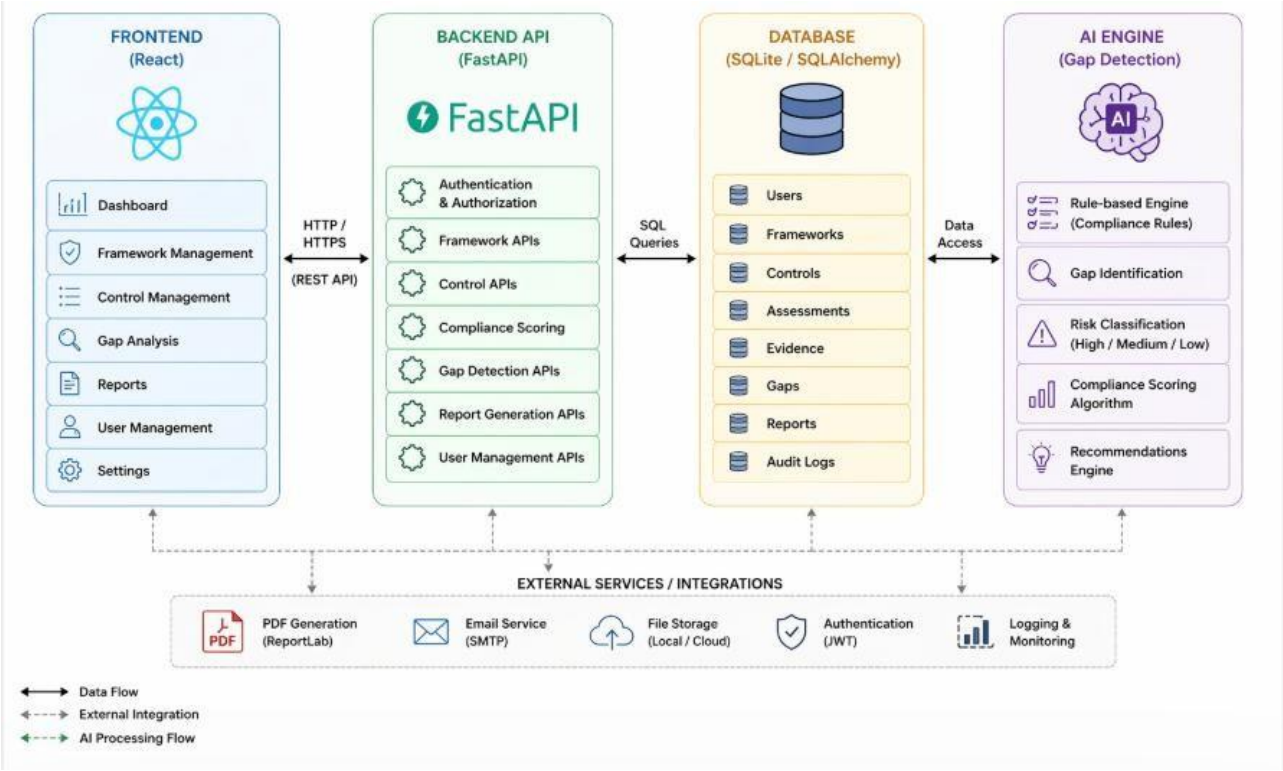

Figure 1.2: System Architecture Overview

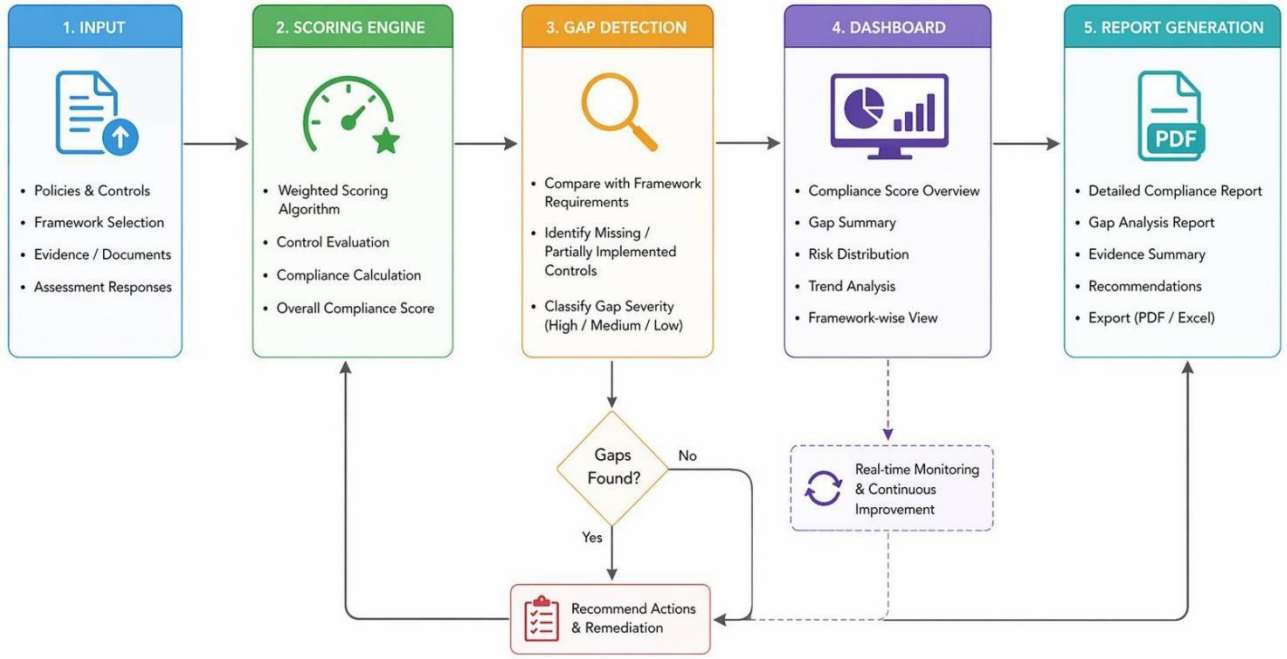

Figure 1.3: Compliance Scoring and Gap Detection Flow

1.5 Objectives

- The main objectives of this project are:
- To design and develop an automated compliance management platform.
- To implement a weighted scoring system for compliance evaluation.
- To develop an AI-based gap detection engine for identifying risks.
- To build a full-stack web application for real-time compliance tracking.
- To generate automated audit reports for improved audit readiness.
- To support multiple compliance frameworks within a single system
- To ensure secure user authentication and role-based access control within the system.
- To provide real-time dashboard visualization for better decision-making and monitoring.

1.6 SDGs (Sustainable Development Goals)

Table 1.3: SDGs and Project Relevance

SDG	Relevance to the Project
SDG 4	This Promotes secure digital infrastructure
SDG 9	Enhances data protection and cybersecurity.
SDG 16	Improves technical education and skill development.
SDG 17	Encourages industry-academia collaboration.

1.7 Overview of the Report

This report is structured into multiple chapters to present the project in a systematic manner: Chapter 1 introduces the background, objectives, and proposed system. Chapter 2 provides a detailed literature survey on compliance systems. Chapter 3 explains the system architecture and technology stack. Chapter 4 covers project management and development phases. Chapter 5 describes system design and implementation. Chapter 6 presents hardware and software requirements. Chapter 7 evaluates system performance and results. Chapter 8 discusses social, legal, and ethical aspects. Chapter 9 highlights learning outcomes. Chapter 10 concludes the project and suggests future enhancements.

Chapter – 2

LITERATURE SURVEY

2.1 Introduction

Compliance management has become a critical component of modern cybersecurity practices due to the increasing number of regulatory frameworks and data protection laws. Organizations are required to follow structured guidelines such as ISO 27001, SOC 2, HIPAA, and DPDP to ensure data security, privacy, and operational integrity. However, traditional compliance processes are largely manual, making them inefficient, time-consuming, and prone to human error.

2.2 Traditional Compliance Management Systems

Traditional compliance management systems rely on manual processes such as spreadsheets, documents, and checklists to track control implementation and audit readiness. These methods lack real-time monitoring, are prone to human errors, and require significant manual effort. As organizations grow and adopt multiple frameworks, managing compliance manually becomes increasingly inefficient and difficult.

2.3 Governance, Risk, and Compliance (GRC) Platforms

GRC platforms such as ServiceNow GRC and RSA Archer provide centralized solutions for managing compliance, risk, and audit workflows. These systems offer automation and reporting capabilities; however, they are often expensive, complex to implement, and require significant configuration. This makes them less suitable for small and medium-sized organizations.

2.4 Modern Compliance Automation Tools

Recent tools like Drata, Vanta, and Secureframe aim to simplify compliance by automating control tracking, evidence collection, and monitoring. These platforms provide dashboards and real-time updates but still rely on predefined logic and lack advanced intelligence for dynamic gap detection and risk prioritization.

2.5 Compliance Scoring Techniques

Compliance scoring is used to evaluate audit readiness. Traditional systems use binary scoring methods, which provide limited insight. Modern approaches use weighted scoring models where different controls are assigned importance based on their criticality. This improves the accuracy of compliance assessment and helps organizations understand their actual compliance posture.

2.6 AI in Compliance and Gap Detection

Artificial Intelligence plays a crucial role in enhancing compliance systems by enabling automated gap detection, risk classification, and remediation recommendations. Rule-based AI systems analyze control data and identify missing or weak controls, classifying them into High, Medium, and Low risk levels. This reduces manual effort and improves decision-making.

2.7 Research Gap Identified

Despite advancements, existing systems lack a unified platform that integrates AI-based gap detection, weighted scoring, multi-framework support, and automated reporting. Many tools are either too complex or lack intelligence. This creates a need for a scalable, intelligent, and cost-effective compliance automation system like the one proposed in this project.

Chapter – 3

SYSTEM ARCHITECTURE & TECHNOLOGY STACK

3.1 System Architecture Overview

The proposed system follows a **3-tier architecture**, consisting of the frontend, backend, and database layers. The frontend (React.js) provides an interactive user interface, the backend (FastAPI) handles business logic and API communication, and the database (SQLite/PostgreSQL) stores compliance data. An additional AI logic layer is integrated to perform compliance scoring and gap detection. This architecture ensures scalability, modularity, and efficient system performance.

3.2 Data Flow

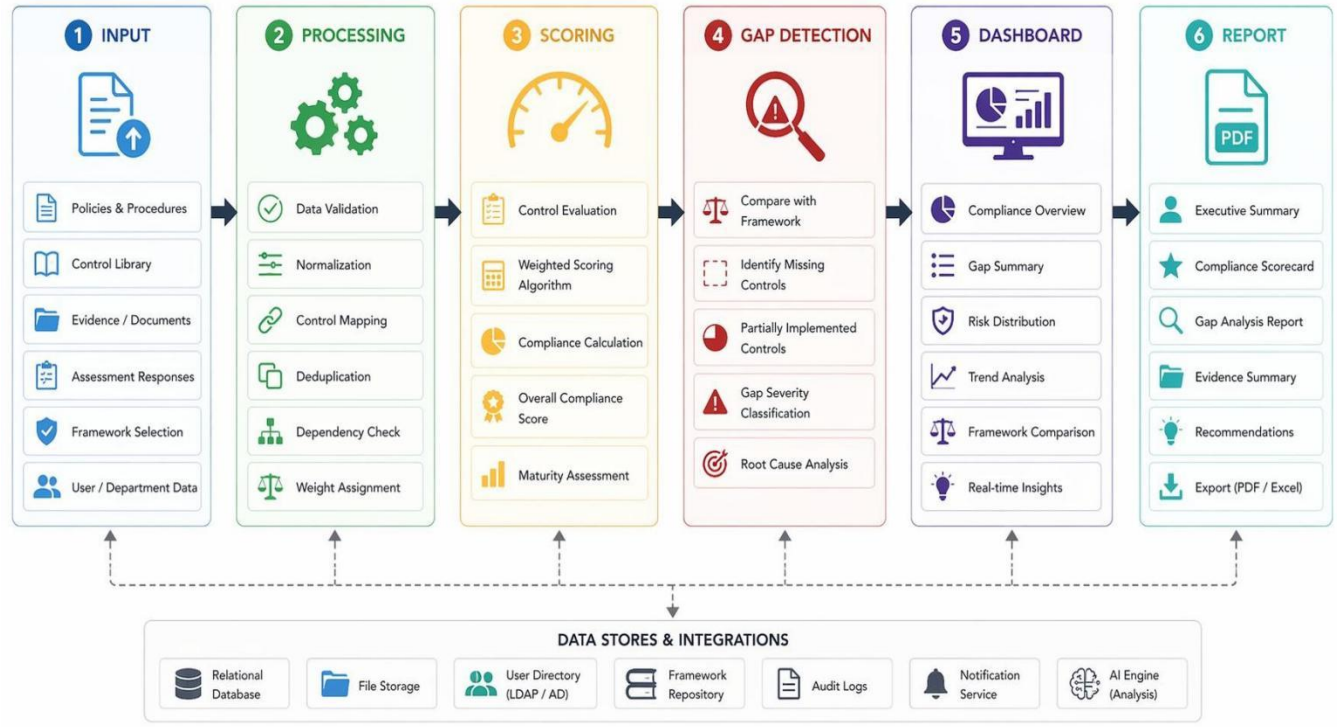

Figure 3.1: End-to-End Data Flow

The system processes data through a structured workflow. Users select a compliance framework and update control statuses. The backend processes this data using a weighted scoring engine and AI-based

gap detection module. The results are displayed on the dashboard, and an automated audit report is generated. This flow ensures real-time updates and efficient compliance tracking.

3.3 Technology Stack

Table 3.1: Technology Stack Summary

Layer	Technology	Version	Purpose
Backend Language	Python	3.10	Server-side logic, AI scoring, data ingestion
Web Framework	FastAPI	0.104.1	REST API, routing, Swagger docs
ASGI Server	Uvicorn	0.24.0	Runs the FastAPI app
ORM	SQLAlchemy	2.0.23	Python-to-database mapping, schema creation
Database	PostgreSQL	15	Relational data storage, 10-table schema
DB Driver	psycopg2-binary	2.9.9	PostgreSQL adapter for SQLAlchemy
Data Validation	Pydantic	2.5.2	Request/response schema validation
JWT Library	python-jose	3.3.0	JWT token creation and decoding
Password Hashing	passlib[bcrypt]	1.7.4	Secure bcrypt password hashing
File Uploads	python-multipart	0.0.6	Multipart form data for CSV uploads
Frontend Framework	React	19.2.4	Single-page application, UI components
Charts	Recharts	3.7.0	Line + bar charts for dashboard
HTTP Client (FE)	Axios	1.13.5	API calls from React to FastAPI
Containerization	Docker + Compose	v3.8	Backend + DB orchestration
CSS Styling	Vanilla CSS	—	Custom dark-theme dashboard styles

3.4 System Modules

The system consists of key modules including Authentication, Control Management, Compliance Scoring, AI Gap Detection, Dashboard, and Audit Report Generation. Each module is designed to handle a specific function, ensuring smooth workflow and modular system design. These modules work together to automate the entire compliance lifecycle efficiently.

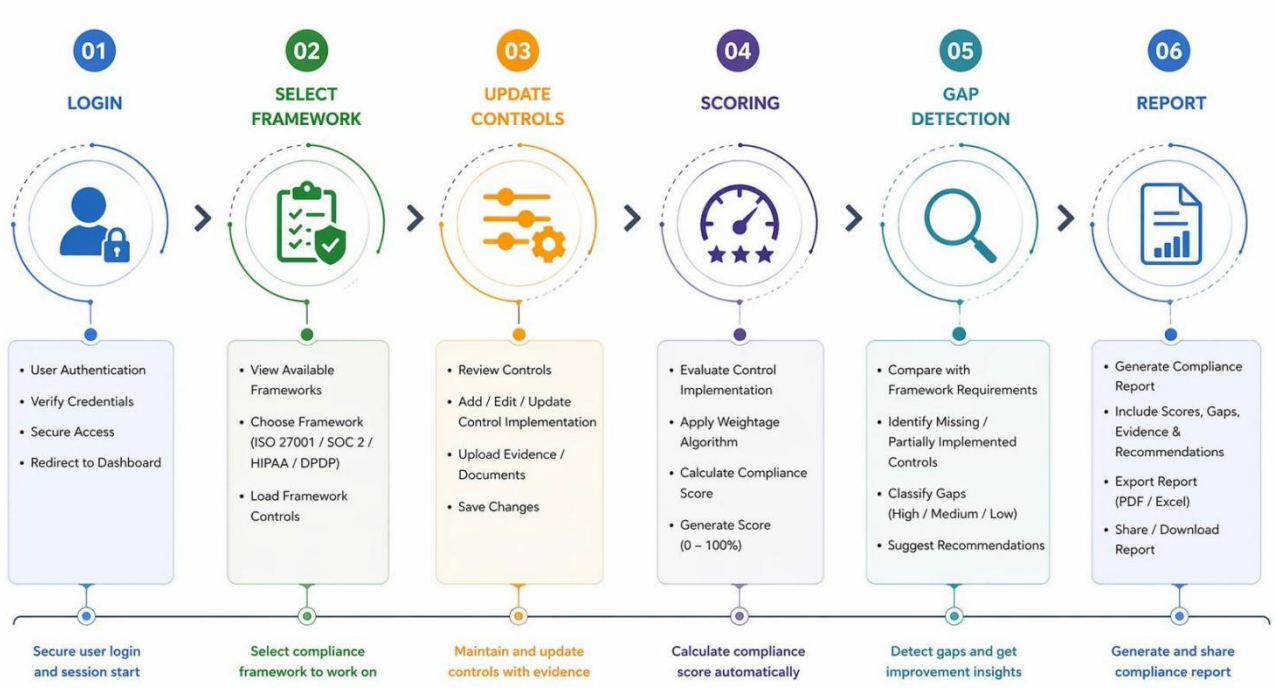

Figure 3.2: System Workflow

Chapter – 4

PROJECT MANAGEMENT

4.1 Internship Timeline

Table 4.1: Internship Timeline

Week	Activities	Deliverables
Week 1	Requirement analysis, framework study, system planning	Project scope, requirement document
Week 2	Database design, control structure setup	Database schema, control mapping
Week 3	Backend development (API + scoring engine)	Working backend APIs
Week 4	Gap detection logic and scoring implementation	AI scoring and gap detection module
Week 5	Frontend dashboard development	UI dashboard with charts
Week 6	Integration of frontend and backend	Fully functional system
Week 7	Testing and bug fixing	Stable system with fixes
Week 8	Documentation and report preparation	Final report and presentation

4.2 Risk Analysis

Table 4.2: Risk Analysis

Risk	Impact	Mitigation
Complex scoring logic	Incorrect compliance results	Use simplified weighted scoring model
Database design issues	Data inconsistency	Use ORM (SQLAlchemy) and validation
API failures	System malfunction	Implement error handling and testing
UI integration issues	Poor user experience	Continuous frontend-backend testing
Data accuracy issues	Incorrect gap detection	Use validated sample dataset

4.3 Internship Budget

Table 4.3: Budget Plan

Item	Estimated Cost (INR)	Remarks
Laptop / workstation usage	0*	Existing personal device used.
Internet access	0*	Used for documentation, Git, and online libraries.
Software tools	0	Open-source Python ecosystem and React libraries.
Development Tools	0*	VS Code, GitHub
Printing / report	150–300	Depending on final submission format.
Miscellaneous	100–250	Stationery and file handling.

** Estimated cost is minimal because the project mainly used existing academic and personal resources.*

Chapter – 5

SYSTEM DESIGN & IMPLEMENTATION

5.1 Compliance Scoring Engine

The Compliance Scoring Engine is the core component of the system responsible for calculating the overall compliance percentage. The system uses a **weighted scoring model** based on the implementation status of each control.

- Implemented → 100% weight
- Partially Implemented → 50% weight
- Not Implemented → 0% weight

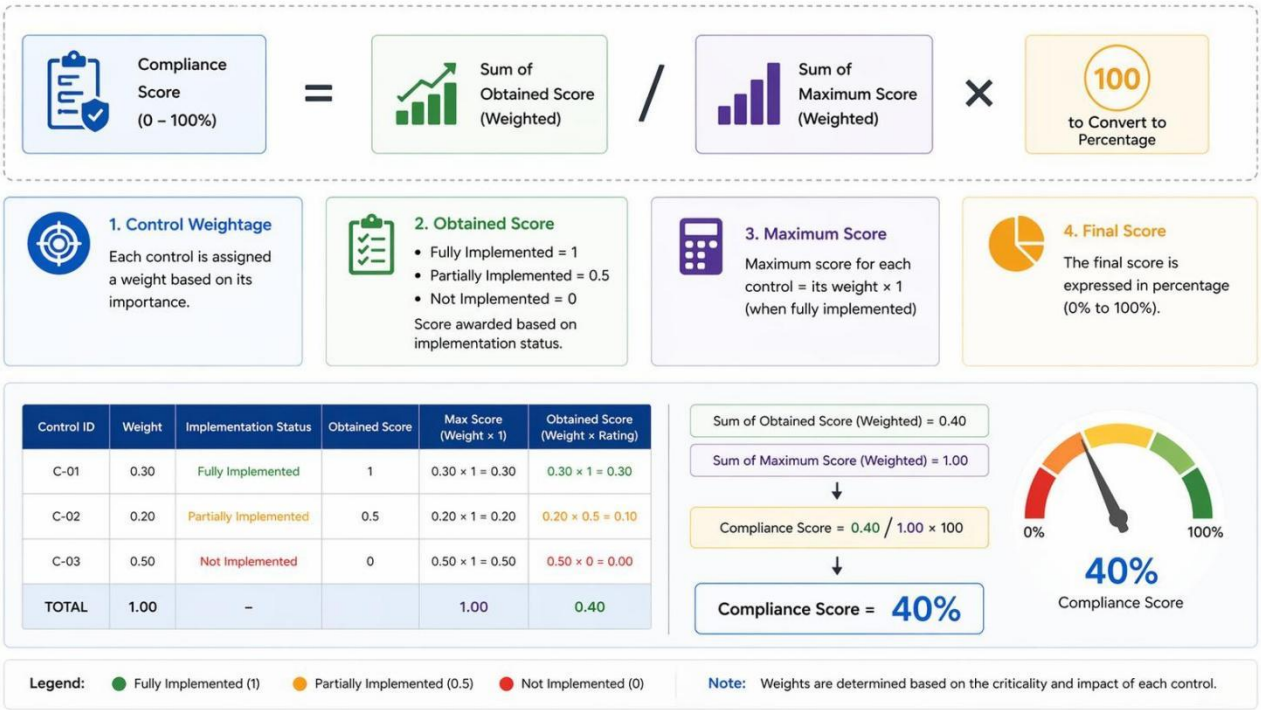

Figure 5.1: Compliance Scoring Formula

The final compliance score is calculated as:

Compliance Score = (Achieved Score ÷ Total Possible Score) × 100

This approach provides a more accurate and realistic evaluation of compliance readiness compared to traditional binary scoring methods.

The system includes a rule-based AI engine that automatically identifies compliance gaps. It analyzes the control status and categorizes each control into risk levels. The system also generates automated remediation suggestions for each identified gap, helping organizations take corrective actions efficiently.

The compliance score represents the overall level of adherence to selected frameworks based on control implementation status. It is calculated using a weighted scoring approach where implemented controls contribute full value, partially implemented controls contribute partial value, and non-implemented controls contribute zero. This scoring mechanism provides a clear and quantitative measure of compliance readiness.

High Risk → Not Implemented critical controls

Medium Risk → Partially Implemented controls

Low Risk → Minor or low-impact controls

Compliance Level	Score Range (%)	Control Implementation Status	Color Code
Critical	80 – 100	Not Implemented (Critical Controls Missing)	High Risk
High	60 – 79	Partially Implemented (Important Controls Incomplete)	Medium Risk
Medium	40 – 59	Moderately Implemented (Some Gaps Present)	Low to Medium Risk
Low	20 – 39	Mostly Implemented (Minor Issues)	Low Risk
Compliant	0 – 19	Fully Implemented (No Significant Gaps)	Very Low Risk

Compliance Score Formula

$$\text{Compliance Score} = \frac{\text{Achieved Score}}{\text{Total Possible Score}} \times 100$$

This approach provides a more accurate and realistic evaluation of compliance readiness compared to traditional binary scoring methods.

Figure 5.2: Compliance Risk Classification and Scoring Model

5.2 Database Design

Table 5.2: Database Tables and Their Purpose

Table	Purpose
Users	Stores user credentials and roles
Frameworks	Stores compliance frameworks (ISO, SOC 2, etc.)
Controls	Stores all compliance controls
Control Status	Tracks implementation status
Risk Assessments	Stores risk classification
Audit Reports	Stores generated reports

5.3 API Endpoints

Table 5.3: API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/auth/register	Register a new user with bcrypt-hashed password
POST	/api/v1/auth/login	Login and receive JWT bearer token (8hr expiry)
GET	/api/v1/ingest-frameworks	Fetch frameworks
GET	/api/v1/controls	Fetch controls
POST	/api/v1/update-status	Update control status
GET	/api/v1/dashboard	Get compliance metrics
GET	/api/v1/report	Generate audit report

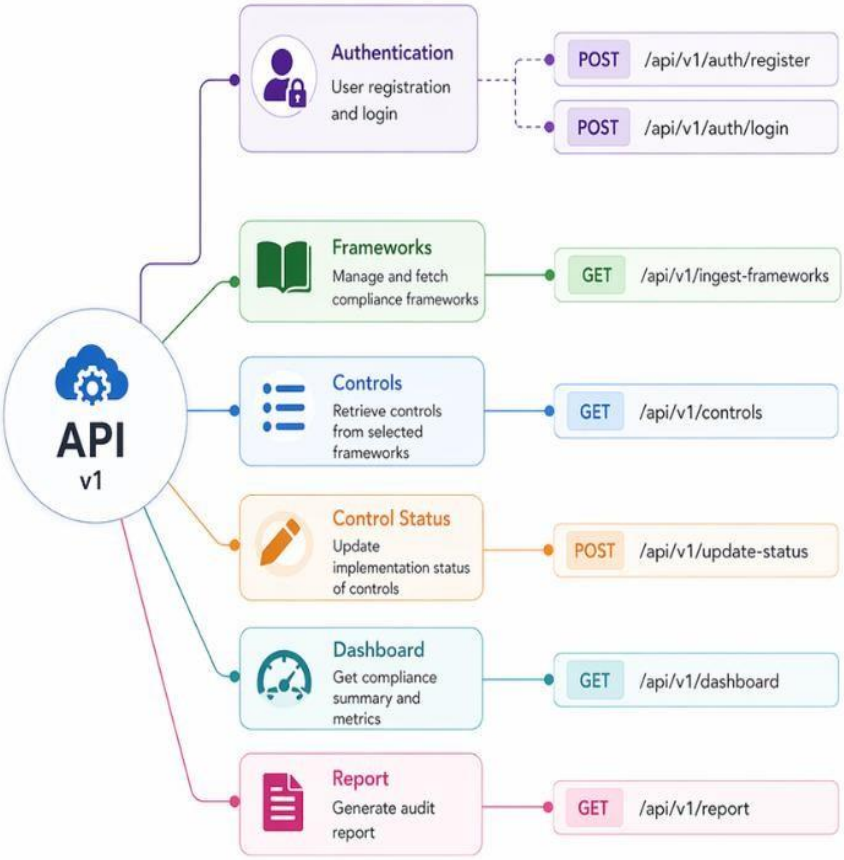

Figure 5.3: API Endpoints Architecture Diagram

5.4 Frontend Dashboard

The frontend dashboard provides a user-friendly interface for monitoring compliance status, displaying key insights such as compliance score visualization, control status distribution charts, risk classification charts, and framework-wise comparisons. It updates in real-time as control data changes, enabling users to gain immediate and accurate insights into compliance performance. Additionally, the system includes an automated audit report generation module using ReportLab, which produces structured PDF reports containing an executive summary, compliance score, gap analysis, risk classification, and remediation plan. This functionality significantly reduces manual effort, ensures consistency in documentation, and improves overall audit readiness.

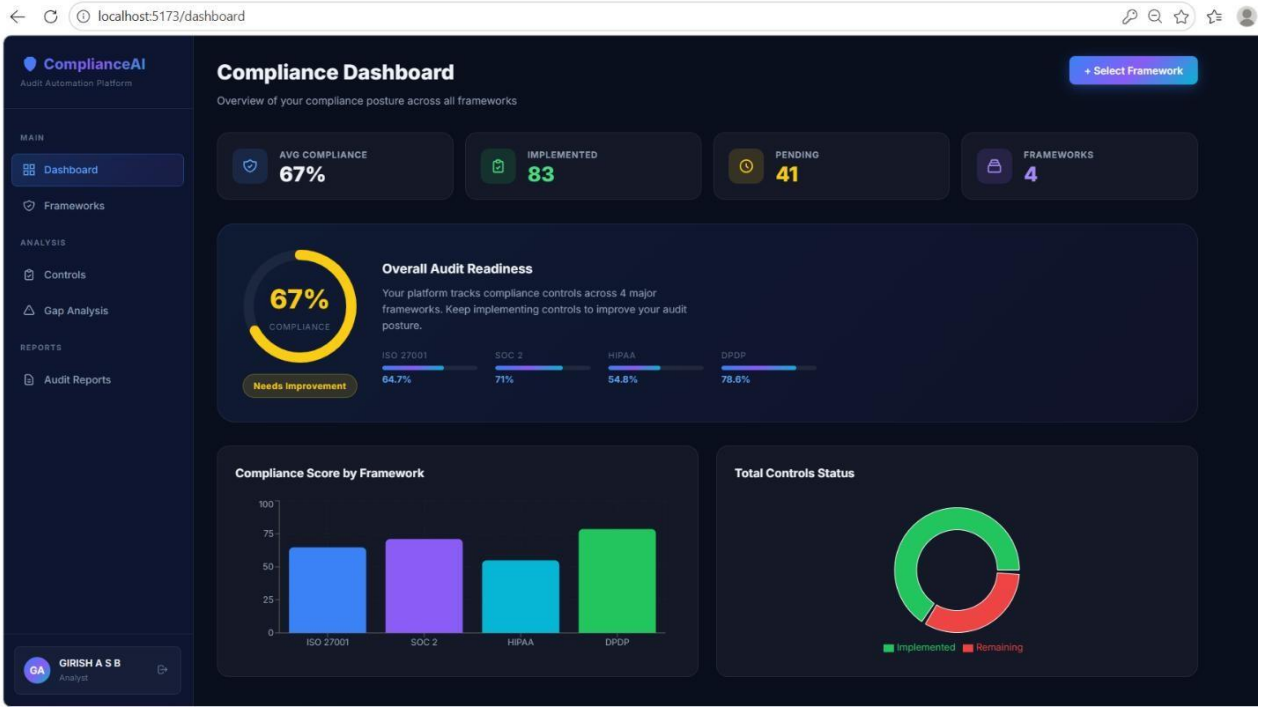

Figure 5.4: Compliance Dashboard with KPI Metrics and Charts

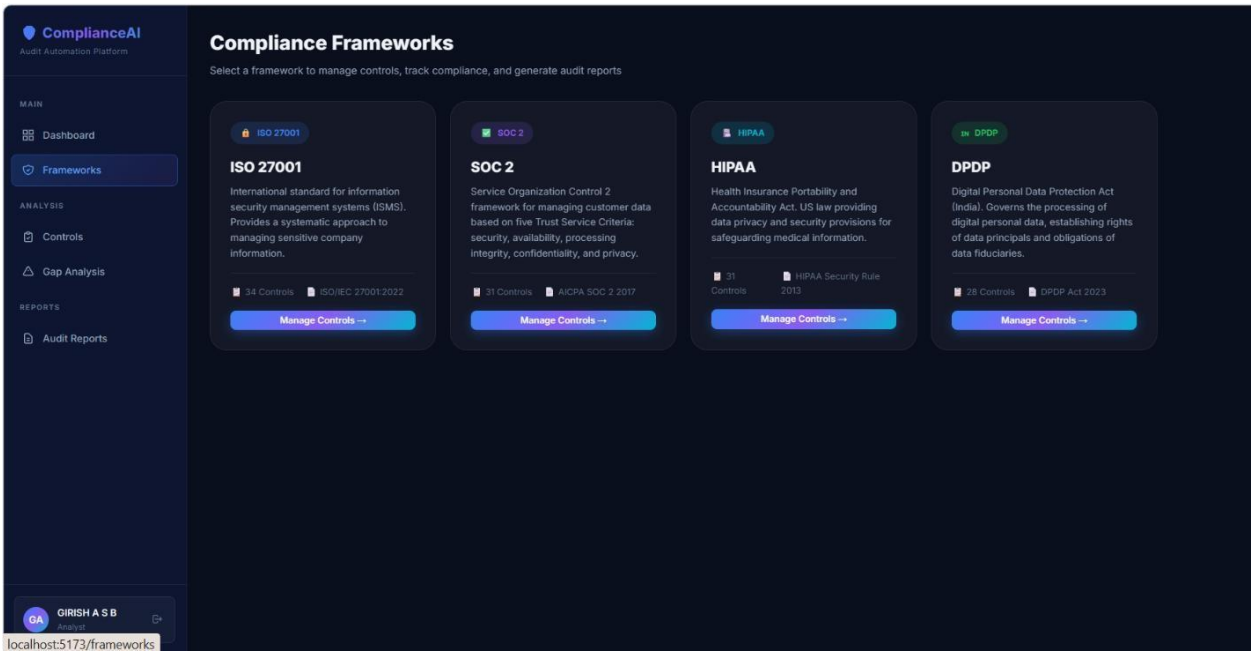

Figure 5.5: Compliance Framework Selection Interface

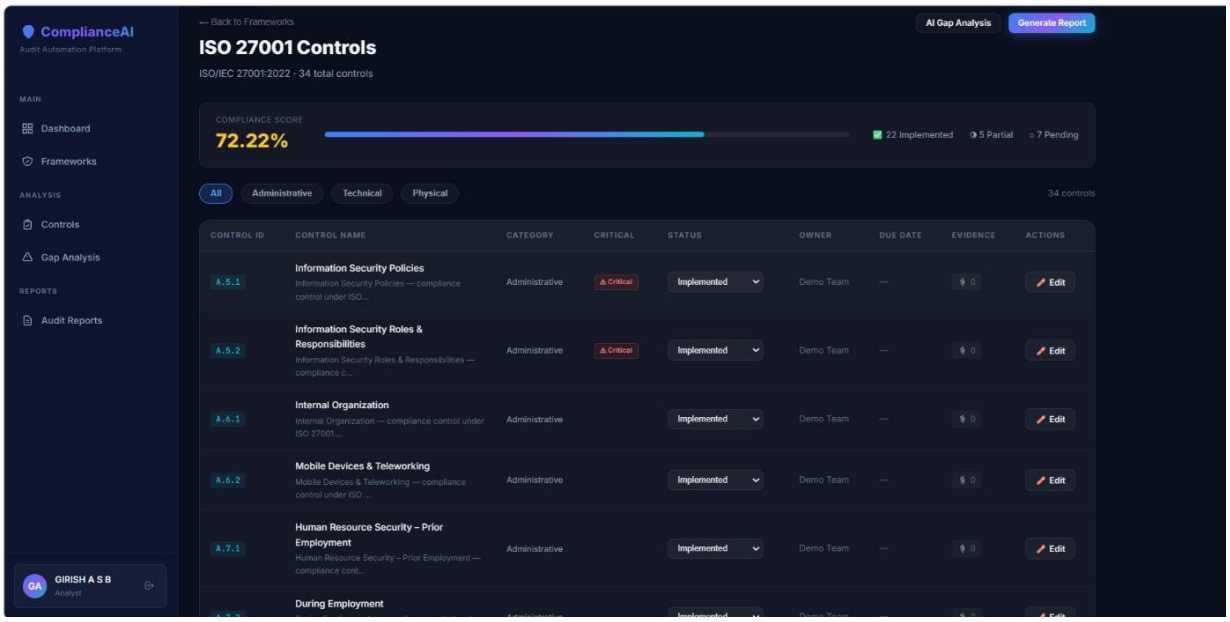

Figure 5.6: Control Management and Compliance Tracking

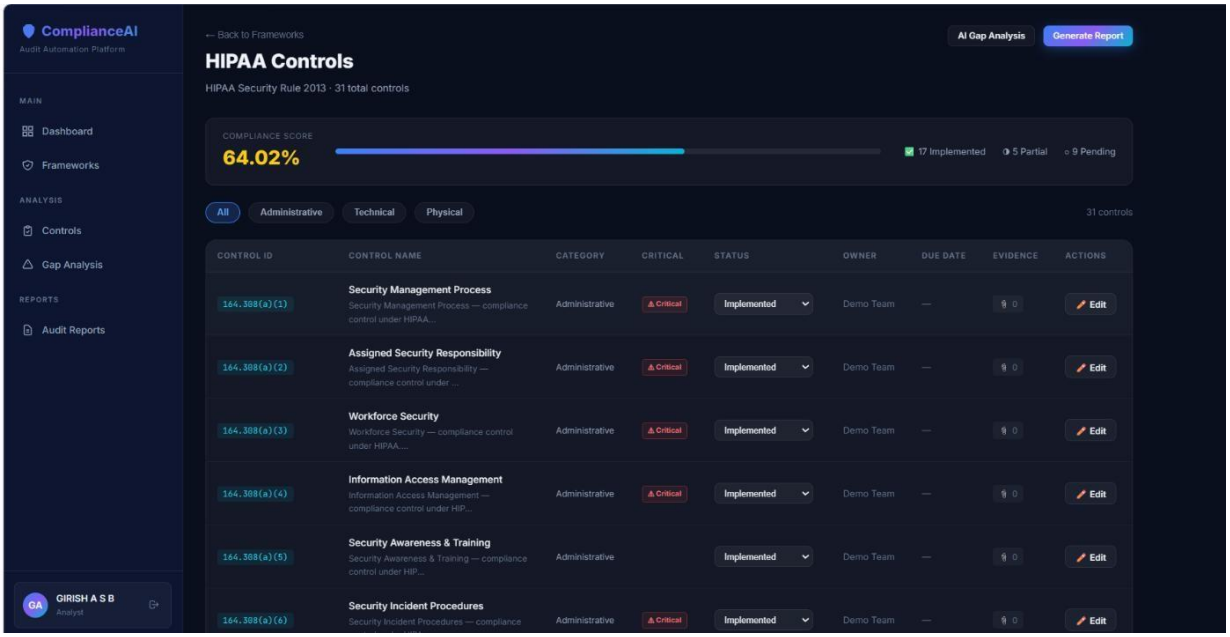

Figure 5.7: Framework Control Analysis and Status View

Chapter – 6

HARDWARE, SOFTWARE AND SIMULATION

6.1 Hardware

The system requires a standard computing environment capable of supporting full-stack web development and database operations. Since the platform is a web-based application, the hardware requirements are moderate.

Table 6.1:Hardware Requirements

Component	Specification	Purpose
Processor	Intel Core i5 or equivalent	Backend processing
RAM	8 GB minimum	Running frontend, backend, database
Storage	20 GB free space	Project files and database
Network	Standard internet connection	API communication and updates
OS	Windows 10/11 or Linux	Development and deployment

6.2 Software Development Tools

Table 6.2: Software Tools and Their Purpose

Tool / Library	Purpose
Python 3.10	Backend development
FastAPI	API development
React.js	Frontend interface

SQLite/PostgreSQL	Database management
SQLAlchemy	ORM for database
Axios	API communication
Recharts	Data visualization
ReportLab	PDF report generation
VS Code	Code editor
Git & GitHub	Version Control

6.3 Simulation and Testing

The system was tested using a simulated dataset consisting of 124 compliance controls across 4 frameworks. Multiple test cases were executed to validate scoring accuracy, gap detection, dashboard updates, and report generation. The simulation ensured that Compliance scores were calculated correctly, Gap detection accurately classified risks, Dashboard displayed real-time updates and Audit reports were generated successfully.

The testing process confirmed that the weighted scoring engine correctly calculated compliance scores, while the gap detection module accurately identified and classified risks into High, Medium, and Low categories. The dashboard displayed real-time updates based on control changes, and the system successfully generated structured audit reports. Overall, the results demonstrated that the system is reliable, efficient, and capable of supporting effective compliance management.

Multiple test cases were designed to validate key functionalities such as compliance scoring, gap detection, dashboard updates, and report generation. Different control statuses, including Implemented, Partially Implemented, and Not Implemented, were used to simulate real-world scenarios and ensure accurate system behaviour.

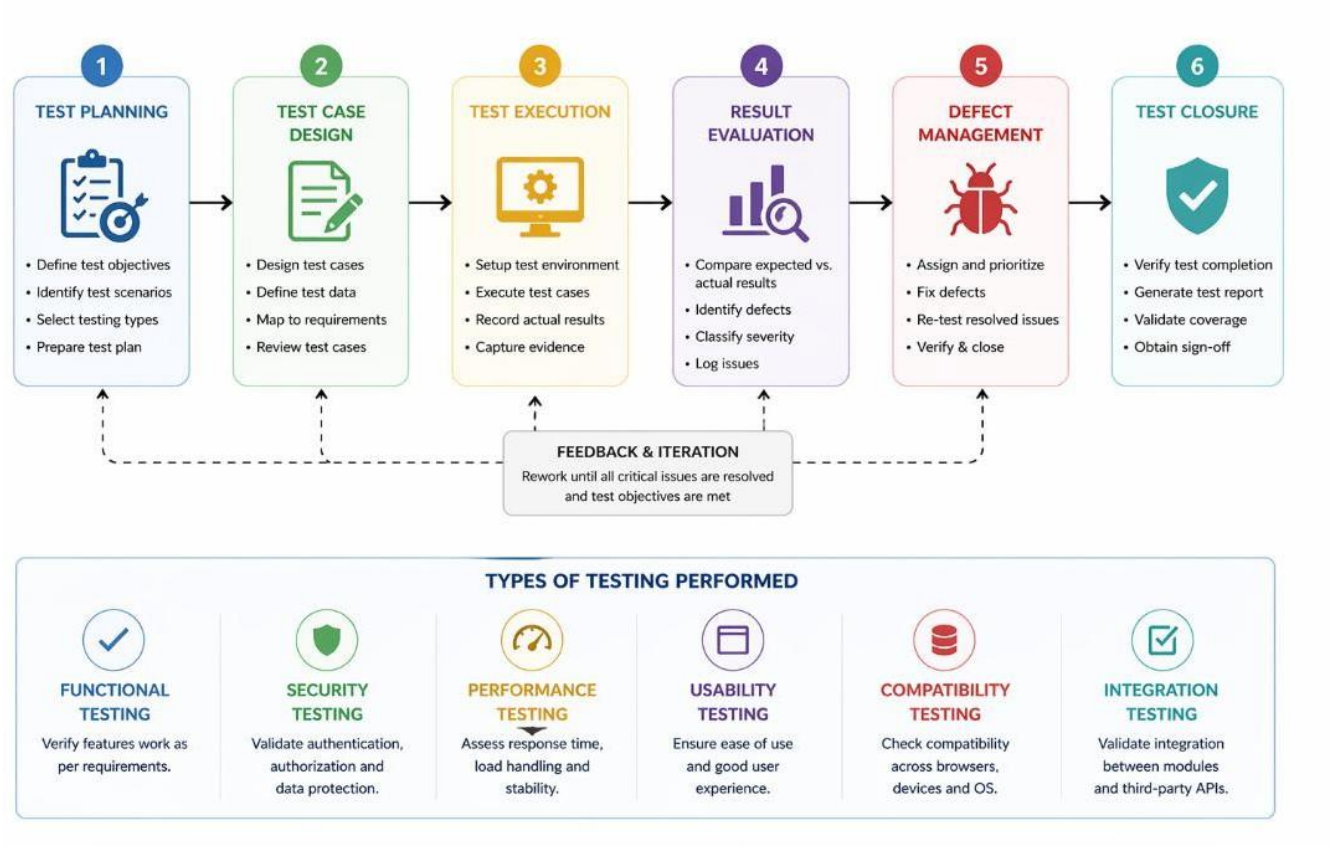

Figure 6.1: System Testing Workflow Diagram

Chapter – 7

EVALUATION AND RESULTS

7.1 Introduction

This chapter presents the evaluation of the AI-Based Compliance and Audit Automation Platform. The system was tested using a simulated dataset to validate its functionality, accuracy, and performance. The evaluation focuses on compliance scoring, gap detection, dashboard visualization, and audit report generation.

7.2 Dataset Description

The system was evaluated using a structured dataset consisting of multiple compliance frameworks and control records. This dataset provides a realistic environment for testing compliance automation features.

Table 7.1: Data Summary

Parameter	Value
Number of Frameworks	4 (ISO 27001, SOC 2, HIPAA, DPDP)
Total Controls	124
Control Status Records	Hundreds of simulated entries
Risk Categories	High, Medium, Low
CVE-2023-2005	Firewall-01

7.3 Results and Observations

The system successfully processed all control data and generated accurate compliance scores. The dashboard displayed real-time updates, reflecting changes in control status instantly. Key observations include Accurate calculation of compliance percentage, Effective identification of missing and weak

controls, Real-time visualization of compliance metrics and Smooth generation of audit reports. The system demonstrated stability and responsiveness during testing.

7.4 Performance Analysis

The platform showed efficient performance in handling multiple frameworks and control records. The backend processed scoring and gap detection quickly, while the frontend provided seamless user interaction.

Table 7.2: System Performance

Feature	Result
Compliance Scoring	Accurate and consistent
Gap Detection	Correct risk classification
Dashboard Update	Real-time
Report Generation	Successful
System Stability	High

7.5 Audit Report Analysis

The system generated structured PDF audit reports automatically. The generated report clearly highlighted compliance gaps and provided actionable recommendations, improving audit readiness.

The reports included:

- Executive summary
- Compliance score
- Gap analysis
- Risk classification
- Remediation plan

The audit report generation module was analyzed to evaluate its accuracy, completeness, and usability. The system successfully generated structured PDF reports containing key components such as the executive summary, overall compliance score, gap analysis, risk classification, and recommended remediation actions. The reports were consistent, well-organized, and provided clear insights into compliance status, enabling stakeholders to easily understand critical issues and take appropriate actions. This automated approach significantly reduces manual effort, ensures standardized documentation, and enhances audit readiness for organizations.

AI-Based Compliance Audit Report

Framework: SOC 2 | Generated: 2026-03-09 17:47 UTC

1. Executive Summary

This audit report presents the compliance posture of the organization against the **SOC 2** framework. The overall compliance score is **0.0%**, indicating **Critical – Not Ready** status. Out of 31 controls assessed, 0 are fully implemented, 0 are partially implemented, and 31 have not been implemented.

2. Compliance Score Summary

Metric	Value
Compliance Score	0.0%
Total Controls	31
Implemented	0
Partially Implemented	0
Not Implemented	31
High-Risk Gaps	15
Audit Readiness	Critical – Not Ready

3. Gap Analysis

Control ID	Name	Category	Status	Severity
CC1.1	COSO Principle 1 – Integrity & Ethi	Administrative	Not Implemented	High
CC1.2	Board Independence & Oversight	Administrative	Not Implemented	High
CC1.3	Organizational Structure	Administrative	Not Implemented	Medium
CC1.4	Competence & Training	Administrative	Not Implemented	Low
CC1.5	Accountability	Administrative	Not Implemented	Low
CC2.1	Risk Communication	Administrative	Not Implemented	Low

Figure 7.1: Automated Audit Report with Compliance Score and Gap Analysis

7.6 Key Observations

The system significantly reduces manual effort in compliance tracking. Automated scoring improves accuracy and consistency. AI-based gap detection enhances decision-making. Real-time dashboard improves visibility. Automated reporting simplifies audit preparation

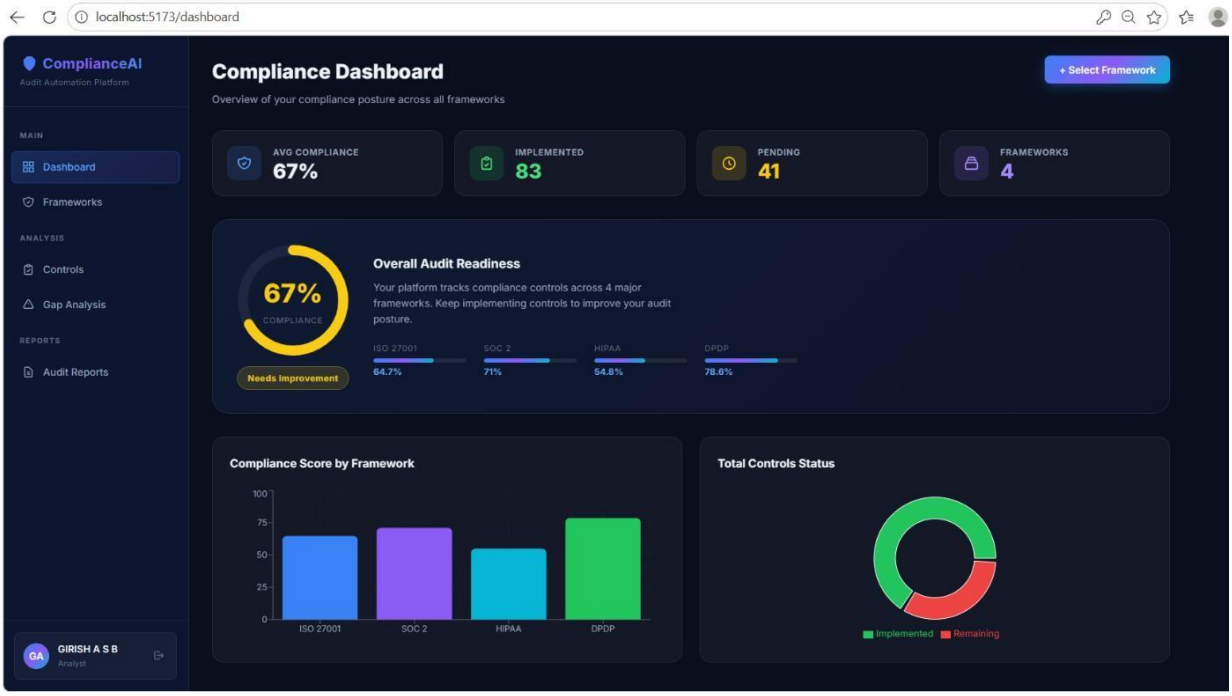

Figure 7.1: Compliance Dashboard with KPI Metrics and Charts

Chapter – 8

SOCIAL, LEGAL AND ETHICAL ASPECTS

8.1 Introduction

The development of an AI-Based Compliance and Audit Automation Platform involves handling sensitive organizational data, including compliance records, control status, and risk assessments. Such systems must consider social impact, legal regulations, and ethical responsibilities to ensure safe and responsible usage. This chapter discusses how the proposed system aligns with these aspects and maintains security, transparency, and accountability.

8.2 Social Impact

The system has a positive social impact by improving organizational cybersecurity and reducing the risk of data breaches. By automating compliance processes, it helps organizations maintain stronger security practices, thereby protecting sensitive user data and preventing cyber threats.

Additionally, the platform benefits small and medium-sized enterprises (SMEs) by providing a cost-effective solution for compliance management. It reduces dependency on manual processes and enables organizations to maintain compliance efficiently, contributing to a safer digital ecosystem.

8.3 Legal Considerations

The system must comply with relevant data protection laws and cybersecurity regulations. In India, the Information Technology Act, 2000 governs data security and privacy requirements. The platform also aligns with global standards such as GDPR, ensuring proper handling of sensitive information.

Legal considerations include Secure storage and processing of compliance data. Restricted access to authorized users only, Protection against unauthorized data exposure, Compliance with audit and regulatory standards. Ensuring legal compliance enhances the reliability and trustworthiness of the system.

8.4 Ethical Considerations

Ethical aspects of the system focus on transparency, fairness, and responsible use of AI. The platform uses a **rule-based AI system**, ensuring that decisions such as compliance scoring and gap detection are explainable and understandable.

Key ethical considerations include Transparency in scoring and risk classification, Avoidance of biased or misleading results, Ensuring data confidentiality, Maintaining accountability in decision-making. The system is designed to support decision-making rather than replace human judgment, ensuring ethical and responsible use.

8.5 Data Privacy and Security

The system handles sensitive compliance data such as control status, risk assessments, and audit reports, which must be protected from unauthorized access. To ensure data privacy and security, the platform implements secure authentication mechanisms such as JWT-based login and password hashing. Access to system features is restricted based on user roles, ensuring that only authorized users can view or modify data.

In addition, the system follows best practices such as secure API communication, input validation, and protection against common vulnerabilities. Proper data handling and storage mechanisms ensure that sensitive information is not exposed or misused. These measures enhance the overall security of the platform and maintain user trust.

Chapter – 9

LEARNING OUTCOMES

9.1 Introduction

This chapter presents the knowledge and skills gained during the development of the AI-Based Compliance and Audit Automation Platform. The internship provided practical exposure to cybersecurity concepts, full-stack development, and system design, bridging the gap between theoretical learning and real-world application.

9.2 Technical Knowledge Gained

The project provided a strong understanding of compliance frameworks such as ISO 27001, SOC 2, HIPAA, and DPDP. It also enhanced knowledge in cybersecurity practices, risk assessment, and compliance management.

From a technical perspective, the development process improved understanding of backend technologies such as FastAPI, API design, database management using SQLAlchemy, and frontend development using React.js. The implementation of a weighted scoring engine and AI-based gap detection strengthened concepts related to system logic and decision-making.

9.3 Practical Skills Developed

The internship helped in developing practical skills in building a complete full-stack application. It involved designing database schemas, implementing REST APIs, integrating frontend and backend systems, and handling real-time data updates.

Additionally, the use of tools such as Git and GitHub improved version control practices, while debugging and testing enhanced problem-solving abilities. The experience of generating automated reports and dashboards provided exposure to real-world software development workflows.

9.4 Analytical and Problem-Solving Skills

Designing the compliance scoring engine and gap detection logic required analytical thinking and structured problem-solving. Challenges such as handling multiple frameworks, ensuring accurate scoring, and maintaining system efficiency were addressed through systematic approaches.

Troubleshooting issues related to API integration, data consistency, and UI updates improved logical thinking and debugging skills. The project strengthened the ability to analyze problems and implement effective solutions.

9.5 Professional and Communication Skills

The internship improved professional skills such as documentation, reporting, and communication. Preparing the project report and presenting the system enhanced the ability to explain technical concepts clearly. Interaction with mentors and continuous feedback helped in understanding industry expectations and improving work quality. The experience also developed time management and project planning skills.

9.6 Overall Learning Outcome

Overall, the internship provided a comprehensive learning experience in cybersecurity, software development, and AI-based system design. It enhanced both technical and professional skills, preparing for future roles in software engineering and cybersecurity domains. The project demonstrated the ability to design and implement a complete system that addresses real-world challenges.

Chapter – 10

CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

This internship project successfully designed and developed an AI-Based Compliance and Audit Automation Platform that automates compliance tracking, gap detection, and audit report generation. The system overcomes the limitations of traditional manual compliance management by introducing automation, real-time analytics, and intelligent decision-making. The implementation of a weighted compliance scoring engine provides an accurate representation of compliance readiness, while the rule-based AI gap detection module effectively identifies missing or partially implemented controls and classifies them into different risk levels.

The platform supports multiple frameworks such as ISO 27001, SOC 2, HIPAA, and DPDP, enabling organizations to manage compliance within a unified system. The interactive dashboard offers real-time insights into compliance status, and the automated report generation feature simplifies audit preparation by producing structured PDF reports. The evaluation results confirm that the system improves efficiency, reduces manual effort, and enhances accuracy in compliance assessment. Overall, the project presents a scalable, efficient, and practical solution for modern compliance management and audit automation.

10.2 Future Scope

Although the system performs effectively, several enhancements can be implemented to further improve its capabilities. Future developments may include integration with real-time compliance monitoring tools and cloud services, enabling continuous compliance tracking. The use of machine learning models can enhance predictive analysis and improve decision-making. Additionally, implementing a multi-tenant architecture would allow multiple organizations to use the platform efficiently. Features such as automated evidence collection, advanced role-based access control, and mobile application support can further enhance usability and scalability. These improvements can transform the system into a comprehensive enterprise-level compliance management platform.

10.3 System Strengths and Practical Value

The system demonstrates several strengths, including automation of the entire compliance lifecycle, accurate and transparent compliance scoring, and intelligent gap detection with risk classification. The real-time dashboard provides clear visibility into compliance status, while the automated audit report generation simplifies documentation and audit readiness. Furthermore, the support for multiple compliance frameworks within a single platform makes the system highly efficient and adaptable for practical implementation.

10.4 Limitations of the System

Despite its advantages, the system has certain limitations. It currently relies on simulated data rather than real-time integration, which may limit its practical deployment. The use of a rule-based AI system instead of advanced machine learning models restricts predictive capabilities. Additionally, the development environment uses SQLite, which has limited scalability compared to production databases. The system also requires manual input for updating control status, which could be automated in future versions. Addressing these limitations will further enhance system performance and usability.

10.5 Final Summary

In summary, the AI-Based Compliance and Audit Automation Platform successfully demonstrates how automation and intelligent system design can simplify complex compliance management processes. The integration of weighted compliance scoring, AI-based gap detection, real-time dashboard analytics, and automated audit report generation provides a complete solution for improving compliance efficiency and audit readiness.

The project highlights the importance of combining cybersecurity knowledge with modern software development practices to address real-world challenges. By supporting multiple compliance frameworks within a single platform, the system ensures flexibility and scalability for different organizational needs. Although certain limitations exist, the proposed system establishes a strong foundation for future enhancements and advanced implementations.

REFERENCES

[1] ISO/IEC. (2022). *ISO/IEC 27001:2022 — Information Security Management Systems (ISMS) Requirements*. International Organization for Standardization. https://www.iso.org/isoiec-27001-information-security.html

[2] AICPA. (2023). *SOC 2® — System and Organization Controls Reports and Trust Services Criteria*. American Institute of Certified Public Accountants. https://www.aicpa.org

[3] HIPAA Journal. (2023). *Health Insurance Portability and Accountability Act (HIPAA) Compliance Guidelines*. https://www.hipaajournal.com

[4] Government of India. (2023). *Digital Personal Data Protection (DPDP) Act*. Ministry of Electronics and Information Technology. https://www.meity.gov.in.

[5] NIST. (2018). *Cybersecurity Framework (CSF) Version 1.1*. National Institute of Standards and Technology. https://www.nist.gov/cyberframework

[6] CIS Controls. (2023). *CIS Critical Security Controls Version 8 — Control 7: Continuous Vulnerability Management*. Center for Internet Security. https://www.cisecurity.org

[7] OWASP. (2023). *OWASP Top 10:2021 — Web Application Security Risks*. Open Web Application Security Project. https://owasp.org/Top10

[8] FastAPI Documentation. (2024). *FastAPI — Modern Web Framework for Building APIs with Python*. https://fastapi.tiangolo.com

[9] React Team. (2024). *React 18/19 Documentation — Building User Interfaces*. Meta Open Source. https://react.dev

[10] SQLAlchemy Documentation. (2024). *SQLAlchemy 2.0 ORM and Core Documentation*. https://docs.sqlalchemy.org

[11] ReportLab. (2024). *ReportLab PDF Generation Toolkit Documentation*. https://www.reportlab.com

[12] Axios Documentation. (2024). *Axios — Promise-based HTTP Client for JavaScript*. https://axios-http.com

[13] Recharts Documentation. (2024). *Recharts — A Charting Library for React Applications*. https://recharts.org

[14] Python Software Foundation. (2024). *Python 3.10 Documentation*. https://docs.python.org

[15] Tikhomirov, N., & Gordeychik, S. (2022). *Compliance Management in Enterprise Systems: Challenges and Best Practices*. International Conference on Information Security and Cryptography (ICISC), Springer.